SparseFormer: A Credible Dual-CNN Expert-Guided Transformer for Remote Sensing Image Segmentation With Sparse Point Annotation

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***Abstract*— Although significant advances have been made in the semantic segmentation of high-resolution remote sensing (RS) images, obtaining accurate pixelwise annotations remains resource-intensive. We propose SparseFormer, a credible dual-convolutional neural network (CNN) expert-guided Transformer model designed for semantic segmentation using point-level annotations to reduce this annotation burden. SparseFormer comprises three branches, where two CNN branches employ different attention mechanisms to encourage diverse outputs. To enhance the local consistency of pseudolabels, we introduce a pixel-adaptive refinement (PAR) module that dynamically refines CNN output probabilities by incorporating image information during training. A credible assessment is then performed to combine the CNN outputs, producing high-quality pseudolabels that supervise the CNN-Transformer hybrid branch. This hybrid branch integrates global representations with local features, achieving precise segmentation. To further strengthen the CNN branches, we introduce a knowledge distillation strategy that steadily feeds back information from the hybrid branch to CNN branches, mitigating overfitting risks caused by sparse supervision. SparseFormer employs credible assessment to reduce pseudolabel uncertainty, followed by continuous interaction and dynamic information enhancement among the three branches in an end-to-end training process. Extensive experiments on two benchmark datasets demonstrate that SparseFormer significantly outperforms state-of-the-art methods. Our code is available at: https://github.com/Yujia73/SparseFormer.**

***Index Terms*— Knowledge distillation, point annotation, semantic segmentation, Transformer, weakly supervised (WS).**

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I. Introduction

SEMANTIC segmentation is one of the core tasks in the remote sensing (RS) domain, aiming to generate detailed land cover maps by assigning semantic labels to each pixel in an image [1]. These high-resolution maps are crucial in urban studies and resource management [2], [3], [4], [5]. In recent years, due to the powerful feature extraction capabilities of deep learning algorithms, significant advancements have been made in RS image semantic segmentation. However, many existing deep learning methods rely on large amounts of pixel-level annotations for fully supervised (FS) training [6], [7], [8]. The acquisition of such annotations poses numerous challenges, as RS images exhibit more complex shapes and larger scales than natural images [9]. Therefore, the cost of annotating has become a significant barrier for further developing deep learning techniques in RS semantic segmentation.

To address this issue, researchers are gradually exploring semi-supervised learning [2], [10], [11] and weakly supervised (WS) learning [12], [13], [14], [15] methods, aiming to reduce reliance on expensive labeled data while maintaining model performance. WS learning utilizes incomplete or imprecise labels that are easy to obtain, such as image-level, bounding box, scribble, and point annotations [16]. Image-level and bounding box labels are the most cost-effective options, but they struggle to achieve precise object localization due to the complex context relationships among objects in high-resolution RS images. Point annotations are a more user-friendly annotation for annotators who can only mark some interior points of objects without precise boundaries [17]. Although the acquisition cost of point annotations is extremely low, it presents a great challenge for accurate semantic segmentation (Fig. 1).

First, the limited supervised information provided by the point annotations makes it difficult to train accurate segmentation models, resulting in poor boundary distinction [*Challenge 1*; Fig. 1(a)]. Second, high-resolution RS images face interclass homogeneity and intraclass heterogeneity

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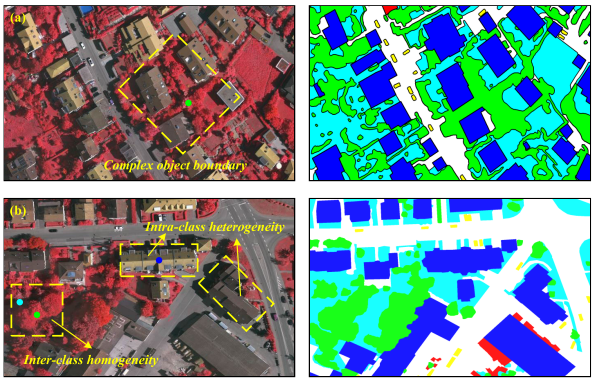


Fig. 1. Challenges of semantic segmentation under point supervised. (a) Challenge 1. (b) Challenge 2.

issues, complicating the ability of the model to distinguish between similar objects [18] under limited information conditions (*Challenge 2*; Fig. 1(b)]. Therefore, there is an urgent need to develop new methods tailored to the characteristics of RS images to enhance the effectiveness of semantic segmentation with sparse point annotations.

Recently, some scholars have tried to use point annotations for the semantic segmentation of RS images. Existing point supervision methods can be categorized into two groups: regularization-based methods [19], [20], [21], [22] and pseudolabel supervision methods [23], [24], [25]. Regularization-based approaches focus on incorporating low-level affinities or high-level semantic affinities derived from the original image to guide training. In contrast, pseudolabel supervision methods aim to generate pseudolabels through sparse annotations and subsequently leverage these pseudolabels to train more robust segmentation models. However, pseudolabel-based methods lack effective quality control in the pseudolabel generation, and erroneous pseudolabels may not only degrade the segmentation performance of the model but also increase uncertainty during training. In addition, previous methods are mostly built on traditional convolutional neural networks (CNNs) models, which struggle to effectively capture global information [26], especially when insufficient label information impacts overall performance. In addition, another research that leverages point annotations is known as object counting. These studies primarily focus on using points to count object numbers rather than achieving precise object segmentation. Object counting methods mainly emphasize the spatial distribution of points and generate object counts or density maps through techniques, such as density estimation [27], [28]. In contrast, point-based supervision for semantic segmentation aims to produce complete pixel-level segmentation results.

In other words, improving the quality of pseudolabel generation and enhancing the global learning ability of models with limited information have become potential solutions for further improving the performance of segmentation models. To address *Challenge 1*, we explore how to effectively train high-performance RS image segmentation models under point annotations. Specifically, to tackle the challenge of learning global shape information without complete annotation masks, we investigate the application of Vision Transformers (ViTs) in WS semantic segmentation. ViTs can capture long-range semantic context through multihead self-attention and multilayer perceptrons [29], [30], [31]. CNNs are good at extracting the local details of the image, while Transformers are competent in modeling distant dependencies. Considering that hybrid CNN-ViT architectures present superiority in terms of segmentation performance [32], we design a model that integrates CNNs with the Transformers, allowing the fusion of local and global representations through multiscale interdependencies, thereby achieving superior segmentation performance. To address *Challenge 2*, we propose leveraging the complementary learning capabilities of multiple models to capture the missing information in point annotations. Specifically, we combine CNN models with different attention mechanisms to generate high-confidence pseudolabels under credible assessment, which are then used to train the CNN-Transformer hybrid branch.

Consequently, we introduce SparseFormer, a model designed specifically for the semantic segmentation of RS images with point annotations. SparseFormer has three branches: two CNN branches and a CNN-Transformer hybrid branch. Through a credible evaluation and dynamic fusion of the outputs from the two CNN branches, SparseFormer generates high-quality pseudolabels that guide the training of the hybrid CNN-Transformer branch. Our method enables effective fusion and enhancement of global and local features, ultimately delivering high-precision segmentation results under point supervision. The main contributions of this study are summarized as follows.

1) We propose SparseFormer, a novel method for the semantic segmentation of RS images under point annotations. This method leverages pseudolabels generated by the CNN branches to supervise the hybrid branch, enhancing model training by strictly controlling the quality of these pseudolabels through credible assessment. The hybrid branch combines the local feature extraction capabilities of CNNs with the long-range dependency modeling of ViT, achieving accurate and reliable segmentation results.
2) We devise a dual-CNN branch structure based on different attention mechanisms and a pixel-adaptive refinement (PAR) module is integrated to enhance the local consistency of CNN outputs using image information. Our method encourages the two branches to produce differentiated outputs, which are dynamically mixed based on uncertainty to create high-confidence pseudolabels.
3) We design a hybrid branch that fully utilizes the advantage of CNN in capturing local details and the ability of the Transformer to model long-range dependencies. This hybrid branch is trained using pseudolabels generated by the CNN branches, yielding more accurate segmentation. Moreover, we introduce a knowledge distillation strategy to reinforce the quality of CNN branch outputs during training. This mechanism dynamically enhances the learning capacity of the CNN branches and boosts the overall model performance under sparse point annotations.

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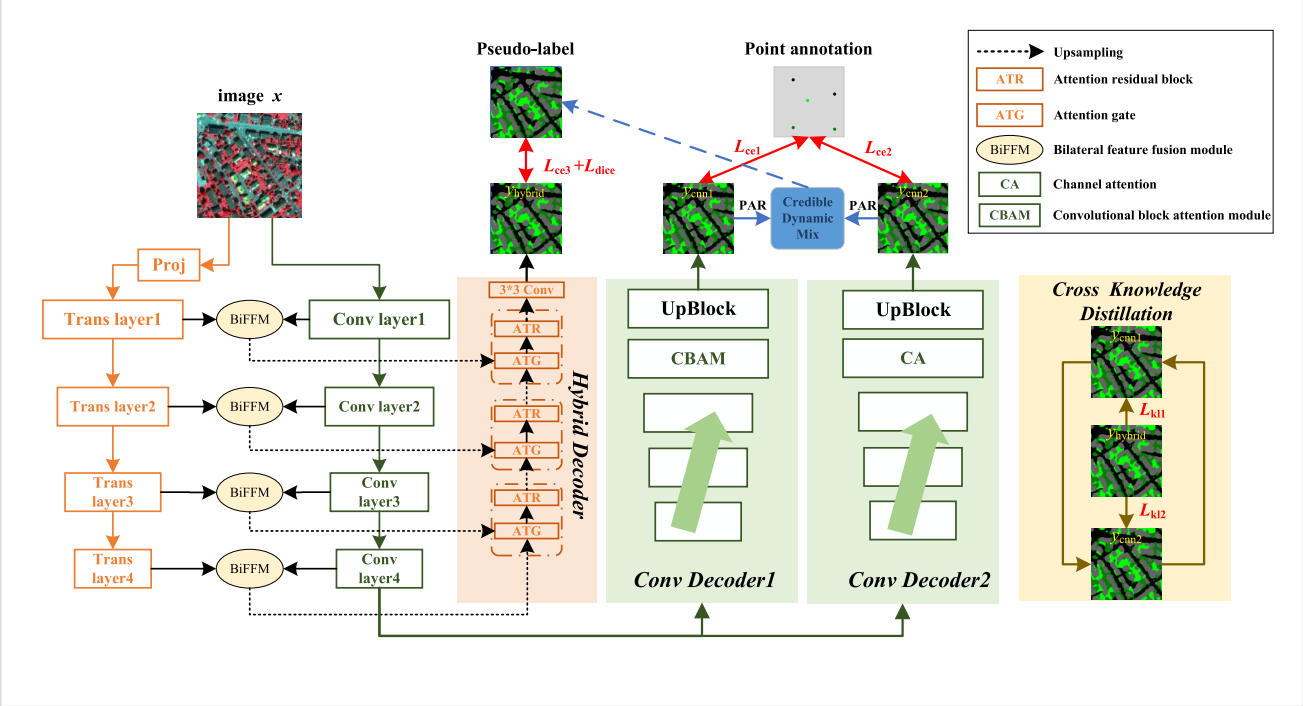


Fig. 2. Workflow of SparseFormer.

4) We validate the proposed method on the Vaihingen and Zurich Summer datasets. The results demonstrate that SparseFormer significantly outperforms existing sparse supervision methods on both datasets, showcasing great potential and advantages in practical applications.

II. Related Works

A. Semantic Segmentation

With the advancement of deep learning, particularly the success of CNNs in image processing, CNNs have gradually become a core technology in the RS semantic segmentation [33], [34]. However, due to the abundant detail in RS images, traditional CNNs face certain limitations in capturing long-range dependencies and handling complex land cover boundaries [34], [35], [36]. Transformer models have recently gained prominence in computer vision to model the global features in RS images better. Transformer models are based on self-attention mechanisms, which enable them to capture long-range dependencies and detailed information in RS images more effectively. Unlike CNNs, Transformers do not rely on a fixed receptive field; instead, they combine global and local feature information through a global self-attention mechanism, thereby enhancing the performance of RS semantic segmentation. For example, Segformer [29], Segmenter [30], and Swin Transformer [37] are commonly used models in the RS semantic segmentation.

However, compared with CNNs, ViTs often overlook local feature details [38]. Therefore, the researchers further proposed a hybrid model combining CNNs and Transformers to use their complementary advantages fully. CNNs take care of extracting the local details of the image, while the Transformer models the distant dependencies. This hybrid architecture has achieved remarkable results in the RS segmentation tasks [26], [39], [40]. This combined approach achieves a better balance in RS semantic segmentation tasks, effectively improving the classification accuracy. Although Transformer-based models have achieved significant success in RS image segmentation, most rely on FS or semi-supervised learning, typically requiring large amounts of fully annotated training data. To the best of the authors' knowledge, Transformer-based techniques have not yet been fully explored for point-supervised RS image segmentation.

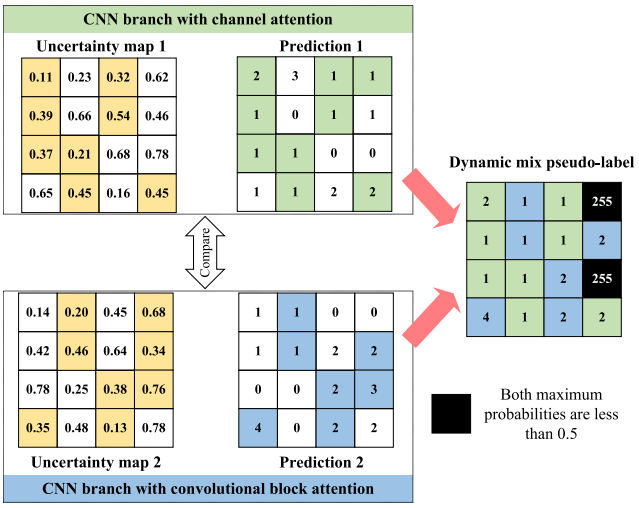


Fig. 3. Dynamic label mixing diagram based on the uncertainty estimation.

B. WS Semantic Segmentation

Common types of WS semantic segmentation methods for RS images can be classified into imprecise supervision, inaccurate supervision, and incomplete supervision [24]. Inaccurate supervision refers to errors or imprecise labeling in the

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annotations, which may mislead the model during training [9], [41]. Imprecise supervision means that the provided label information are not sufficiently precise, such as giving a vague region or imprecise boundaries, but it can still be used for training [42], [43]. Incomplete supervision refers to incomplete label information, where only partial categories or regions are annotated [19], [21], [23], [25], [44], [45]. Among WS semantic segmentation approaches, the point-level supervision is one of the most challenging forms of incomplete supervision.

The existing point-annotation learning methods can be divided into two categories according to how they use the limited information. The first approach focuses on generating pseudolabels from limited annotations, which are then used for training to supplement the missing information. The pseudolabel can somewhat enhance the training data, but its effectiveness heavily relies on the quality of the pseudolabels. For example, Xu and Ghamisi [24] proposed a point-supervised semantic segmentation network based on region growth, which gradually expands the initial sparse annotations labels by iteratively selecting unlabeled regions with high confidence as pseudolabels, thus completing the complete information required for model training. Wu et al. [45] proposed an adaptive Gaussian mixture model (AGMM) sparse supervision semantic segmentation framework that constructs a Gaussian mixture for similar unlabeled pixels, using labeled pixels as centroids to model the feature distribution of each class. This approach supervises the training of unlabeled pixels by leveraging reliable information from labeled pixels and the adaptively generated Gaussian mixture model predictions.

The second type of method primarily enhances or supplements the model learning capability by introducing regularization strategies. For instance, Wu et al. [22] designed a point-supervised deep bilateral filtering semantic segmentation network (DBFNet) that filters deep features through a nonlinear combination of nearby feature values, bringing similar features closer to achieving consistent predictions in smooth areas. Wu et al. [45] proposed a sparse supervised semantic segmentation algorithm based on attention propagation, which provides supervision for unlabeled regions by modeling the relationship between significant regions and the affinity supervision loss function so that objects with similarity in high and low dimensional feature spaces have the same semantics. Hua et al. [20] proposed the feature and spatial relationship regularization method, which supplements the supervised task with an unsupervised learning signal that accounts for both spatial and feature-based neighborhood structures.

We believe that improving the quality of pseudolabels through credible evaluation methods, combined with leveraging Transformers to capture global contextual information, can significantly enhance the segmentation performance of sparse point supervised learning.

III. Method

A. Overviews

To achieve accurate semantic segmentation of RS images under sparse supervision, we propose SparseFormer. This method consists of three main components: high-confidence pseudolabel mining using CNNs with different attention mechanisms, a segmentation head based on a hybrid CNN and Transformer, and a knowledge distillation strategy (Fig. 2). The complete pseudocode of SparseFormer is described in Algorithm 1. Next, we will introduce each component in detail.

B. Dual-CNN Branch Based on Multiple Attentions

The core goal of SparseFormer is to use the CNN outputs as pseudolabels. However, the limited supervised information lead to high uncertainty from the predictions of a single model, making it difficult to generate high-quality pseudolabels. Incorrect pseudolabels will lead the model to optimize in the wrong direction, reducing its performance. To fully leverage the advantages of multiple models, we propose a dual-CNN branch structure based on multiple attention mechanisms to enhance the quality and reliability of pseudolabels. Therefore, we introduce the convolutional block attention module (CBAM) and channel attention (CA) mechanisms [46] in the CNN-based decoder branches. The CA mechanism enhances channel importance and improves feature expressiveness, while the CBAM mechanism focuses on various regions in the image and emphasizes spatial and channel information. This strategy encourages the two branches to produce differentiated outputs, allowing for the integration of diverse perspectives and resulting in more reliable and stable pseudolabels.

The CNN encoder utilizes the lightweight EfficientNet-B3 backbone pretrained on ImageNet [47], extracting shallow features through a 3×3 convolution combined with batch normalization and the Sigmoid activation function. It employs movable invertible convolutional blocks to capture multilevel spatial details while reducing parameters and retaining fine-grained features. For the input image $I \in R^{H\times W\times 3}$, four feature maps $F_i \in R^{(H/2^i)\times(W/2^i)\times C_i}$ are generated, where $C_i \in \{48, 96, 136, 232\}$ and $i \in \{2, 3, 4, 5\}$. The atrous spatial pyramid pooling (ASPP) module uses the output from stage 4, combining dilated convolutional blocks with four different dilation rates, global average pooling, and a 1×1 convolutional block to produce output features. Next, we design two decoders based on different attentions, and they receive low-level feature maps from the intermediate layers and the outputs from the ASPP module. Finally, the two decoders incorporate CBAM and CA mechanisms and compute the cross-entropy loss functions based on point annotations.

C. Credible Dynamic Pseudolabel Mix Module

To generate high-confidence pseudolabels during the training process, we propose a credible dynamic pseudolabel mix module based on the outputs of the CNN branches. By integrating the PAR and credible assessment mechanisms, SparseFormer enhances the quality and reliability of pseudolabels, allowing for real-time adjustments based on the prediction uncertainty. This dynamic approach not only minimizes the impact of noise but also fosters a more robust training process.

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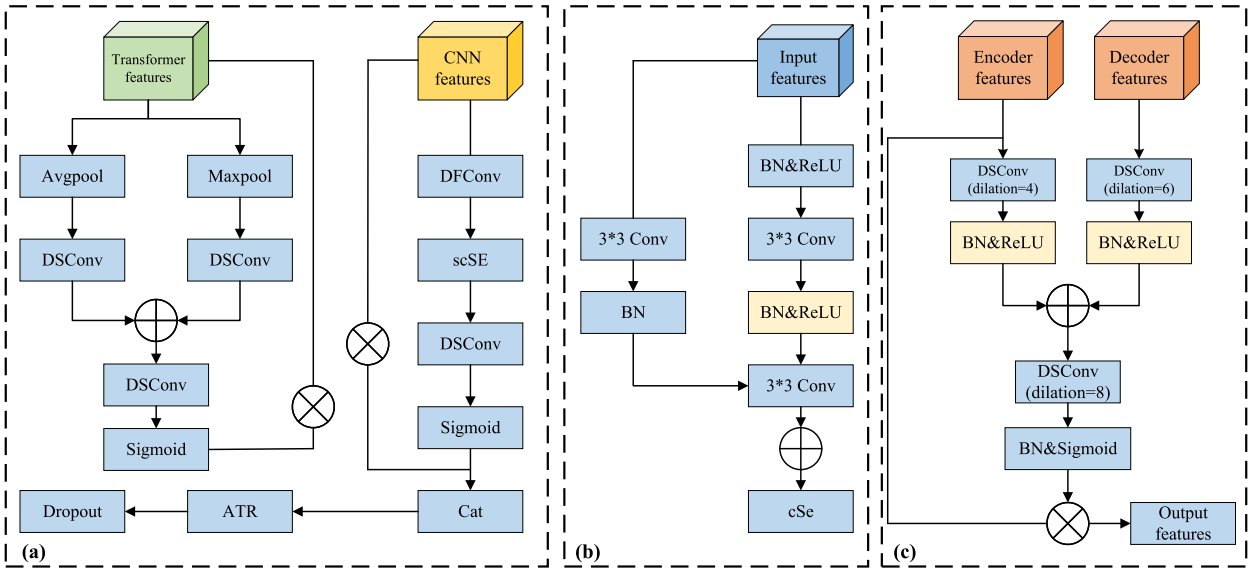


Fig. 4. Schematic of (a) bilateral feature fusion module (BiFFM), (b) ATR, and (c) ATG modules.

1) PAR Module: Due to the inability of point annotations to accurately depict object boundaries, the generated pseudolabels often suffer from poor local consistency. To improve the local consistency, SparseFormer introduces the PAR module [48]. Unlike traditional dense conditional random fields (dCRFs) [49], PAR can adaptively enhance output probabilities end-to-end during model training without high computational costs. By combining color and spatial information, PAR effectively enhances the probability outputs of the two CNN branches, thereby improving pseudolabel quality and overall performance. In addition, the PAR module can also provide extra information, helping to optimize the generalization capability of the model.

Given an input image $I \in R^{H\times W\times 3}$, for the pixel at position (i, j) and (k, l), the RGB and spatial pairwise terms can be expressed as follows:

$$\kappa_{\text{rgb}}^{i,j,kl} = -\left(\frac{\left|I_{ij} - I_{kl}\right|^2}{\omega_1 \sigma_{\text{rgb}}^{i,j}}\right), \quad \kappa_{\text{pos}}^{i,j,kl} = -\left(\frac{\left|P_{ij} - P_{kl}\right|^2}{\omega_2 \sigma_{\text{pos}}^{i,j}}\right) \tag{1}$$

where I_{ij} and P_{ij} denote the RGB values and the spatial coordinates of pixel (i, j), respectively. In practice, the spatial location is defined using the XY coordinates. σ_{rgb} and σ_{pos} refer to the standard deviations of the RGB values and spatial positions. The parameters ω_1 and ω_2 regulate the smoothness of κ_{rgb} and κ_{pos}, respectively. The affinity kernel for PAR is constructed by applying softmax normalization to κ_{rgb} and κ_{pos}, and then summing them as follows:

$$\kappa^{ij,kl} = \frac{\exp\left(\kappa_{\text{rgb}}^{i,j,kl}\right)}{\sum_{(x,y)} \exp\left(\kappa_{\text{rgb}}^{i,j,kl}\right)} + \omega_3 \frac{\exp\left(\kappa_{\text{pos}}^{i,j,kl}\right)}{\sum_{(x,y)} \exp\left(\kappa_{\text{pos}}^{i,j,kl}\right)} \tag{2}$$

where (x, y) is sampled from the neighbor set of (i, j), i.e., $N(i, j)$, and ω_3 adjusts the importance of the position term. Based on the constructed affinity kernel, we refine the initial output of CNN branches. The refinement is conducted for multiple iterations. For output probability $P \in R^{H\times W\times C}$ in iteration t, we have

$$P_t^{i,j,c} = \sum_{(k,l)\in\aleph(i,j)} \kappa^{ij,kl} P_{t-1}^{k,l,c}. \tag{3}$$

For the neighboring pixel sets $\aleph(\cdot)$, we define them as eight-connected neighbors with varying dilation rates. This design enhances training efficiency, as dilated neighbors for a given pixel can be efficiently obtained using 3×3 dilated convolutions.

2) Credible Assessment for Generating Pseudolabels: Due to the limited supervised information available to CNN models, pseudolabels generated by a single model often exhibit significant uncertainty. To overcome this challenge, integrating the uncertainty outputs from two CNN models enables a more systematic identification of high-confidence and low-confidence regions. Hence, we conduct a credible assessment for CNN outputs, ensuring that only high-confidence pseudolabels are utilized for training. Specifically, we first obtain the predicted probabilities P_1^n and P_2^n after enhancement through the PAR module, along with their corresponding prediction masks

$$P_{1-\max}^n = \arg\max P_1^n, \quad P_{2-\max}^n = \arg\max P_2^n \tag{4}$$

where $P_{1-\max}^n \in \mathbb{R}^{H\times W}$ and $P_{2-\max}^n \in \mathbb{R}^{H\times W}$ represent the prediction from P_1^n and P_2^n, respectively.

Next, we estimate the uncertainty of each pixel using entropy. Specifically, we calculate the uncertainty of each pixel using the predicted maximum and second maximum probabilities [9], as shown in the following formula:

$$u_i = -(\hat{y}_{\max} \log \hat{y}_{\max} + \hat{y}_{\text{sec}} \log \hat{y}_{\text{sec}}) \tag{5}$$

where $\hat{y}_{\max}$ and $\hat{y}_{\text{sec}}$ refer to the maximum and second maximum probabilities of one pixel, respectively.

A smaller value of u_i represents less uncertainty, indicating that the corresponding predicted label is more likely to be accurate. After calculating the uncertainty for all pixels, we define an uncertainty map (U) composed of the uncertainty values of all pixels.

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Algorithm 1 Training the Proposed SparseFormer

Input: Training set with sparse point annotations y and image x

Output: Predicted semantic segmentation map

Load pre-trained EfficientNet-B3 weights (ImageNet) for the CNN backbone.
Load pre-trained SwinV2 weights (ImageNet) for the Transformer backbone.
Initialize CBAM, CA, and hybrid decoder parameters with random Gaussian values.
for epoch $\in$ [0, total_epoch) **do**
  Get mini-batch sample (x, y).
  Compute the probability maps for x:
    $P_{\mathrm{CA}} = \sigma(f_{\mathrm{CA}}(x))$, $P_{\mathrm{CBAM}} = \sigma(f_{\mathrm{CBAM}}(x))$, $P_{\mathrm{Hybrid}} = \sigma(f_{\mathrm{Hybrid}}(x))$.
  if epoch $\leq$ warm_epoch **then**
    Compute the segmentation loss L_{seg1} using Equation (11).
  else
    Refine P_{CA} and P_{CBAM} using the PAR module to obtain updated probability maps.
    Generate pseudolabels based on the refined P_{CA} and P_{CBAM} using the credible dynamic mix strategy (Equations (4), (5), (6)).
    Compute the segmentation loss L_{seg2} using Equation (12).
    Feedback information from the hybrid branch and one CNN branch to the other CNN branch using Equations (8), (9), and (10).
  end if
end for
return Final predicted segmentation maps from the Hybrid branch.

Through uncertainty estimation, we obtain two uncertainty maps from the CNN predictions. Then, we devise a dynamic mixing strategy to effectively combine multibranch predictions and reduce the uncertainty of the pseudolabels (see Fig. 3). We determine which CNN branch's prediction to select as the pseudolabel based on confidence comparison. In addition, to mitigate the negative impact of pseudolabel noise on the model, we jointly determine low-confidence regions using the maximum probabilities from both models [15], setting them as background values that do not participate in training. Hence, we can obtain the following reliable predictions:

$$M_{\mathrm{pse}} = \begin{cases} P^n_{1-\max}, & \text{if } u_1 < u_2 \\ P^n_{2-\max}, & \text{if } u_2 < u_1 \\ 255, & \text{if } \hat{y}_{\max 1} < 0.5 \text{ and } \hat{y}_{\max 2} < 0.5. \end{cases} \tag{6}$$

D. Hybrid CNN and Transformer Branch

In SparseFormer, we use two CNN models to generate high-confidence pseudolabels. However, CNNs are limited by their fixed-size local receptive fields, hindering their ability to capture long-distance dependencies and global context. Therefore, we built a hybrid branch trained with dynamic pseudolabels to capture richer semantic information through diverse feature representations and effectively model unlabeled pseudolabel areas by leveraging long-range dependencies. Our method enhances the ability of the model to study complex spatial relationships under incomplete supervision, ultimately improving performance. We introduce the BiFMM module to aggregate semantic features from different levels, the attention gate (ATG) module to capture context within a broad neighborhood, and the attention residual (ATR) module to enhance feature representation using CA [50]. This hybrid branch combines the strengths of CNNs in local feature extraction with those of Transformers in global information modeling, improving overall model performance. Diagrams of the BiFMM, ATR, and ATG modules are shown in Fig. 4.

Specifically, the Transformer encoder uses the SwinV2 backbone pretrained on ImageNet [51]. It first splits the image into nonoverlapping small chunks and then uses a linear embedding layer to project the features of each small chunk into the C-dimension. Then, global context features are extracted using multihead self-attention and shifted windows multihead self-attention (W-MSA). With the increase of the depth of the network, the resolution of the patch decreases with the increase of the number of patch merging layers. For the input image $I \in R^{H\times W\times 3}$, we obtain the characteristics of the four levels of mapping $F_i \in R^{(H/2^i)\times(W/2^i)\times C_i}$, including $C_i \in \{96, 192, 384, 768\}$ and $i \in \{2, 3, 4, 5\}$. In the early stage of training, only sparse labels are used to calculate the cross-entropy loss function of the hybrid branch. After obtaining the high-confidence dynamic mixed pseudolabel M_{pse}, it is fed into the hybrid branch for supervised learning with standard cross entropy and Dice loss functions.

E. Cross-Decoder Knowledge Distillation

The hybrid branch learns richer and more comprehensive feature representations by integrating the strengths of CNNs and Transformers. In contrast, the CNN branch that solely relies on initial point annotations is limited in its structure and expressive capability during the learning process, resulting in constrained quality of generated pseudolabels. To further reinforce the performance of the CNN branch, we provided feedback on multilevel and multiscale features learned from the hybrid branch and another CNN branch to the remaining CNN branch. This information feedback mechanism helps the CNN branch better capture global and local features, improving its understanding of complex scenes and thus generating more accurate and high-quality pseudolabels. To achieve this, we devise a cross-decoder knowledge distillation module, which effectively stimulates feature complementarity among different branches. This feedback mechanism significantly enhances the CNN branch's ability to learn from point annotations, enabling it to generate more efficient pseudolabels by leveraging the advantages of the hybrid branch, thereby improving the overall performance and accuracy. We follow the practice of knowledge distillation [52] and employ temperature-calibrated Softmax (T-Softmax) to

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soften the probability maps

$$\tilde{p}_c = \frac{\exp\left(\frac{z_c}{T}\right)}{\sum_c \exp\left(\frac{z_c}{T}\right)} \tag{7}$$

where z represents the logit prediction of pixel class c, $\tilde{p}_c$ represents the soft probability value of class c, and temperature T is the parameter controlling the output probability softness. $T = 1$ corresponds to a standard Softmax function, the higher the value, the softer the output probability distribution and the higher the entropy.

Supposing $\tilde{p}_{\text{CBAM}}$, $\tilde{p}_{\text{CA}}$, and $\tilde{p}_{\text{Hybrid}}$ are the T-Softmax soft probabilities of two CNN branches and hybrid branches, respectively. The knowledge distillation loss of two CNN branches is

$$L_{\text{kd}}^{\text{CBAM}} = \text{KL}(\tilde{p}_{\text{CBAM}}, (\tilde{p}_{\text{CA}} + \tilde{p}_{\text{Hybrid}})/2) \tag{8}$$

$$L_{\text{kd}}^{\text{CA}} = \text{KL}(\tilde{p}_{\text{CA}}, (\tilde{p}_{\text{CBAM}} + \tilde{p}_{\text{Hybrid}})/2) \tag{9}$$

where KL$(\cdot)$ represents the Kullback–Leibler divergence loss function [53].

Therefore, the total knowledge distillation loss function is

$$L_{\text{kd}} = L_{\text{kd}}^{\text{CBAM}} + L_{\text{kd}}^{\text{CA}}. \tag{10}$$

F. Joint Loss Function

SparseFormer trains from end to end. Since the model could not make reliable predictions at the initial stage of training, we first designed the warm-up stage, which only conducted the training of sparse label supervised loss function for ten epochs. Supposing the outputs of the CA, CBAM, and hybrid branches of SparseFormer are: y_i^1, y_i^2, and y_i^3, with the initial point annotation M_{initial}. The loss function of the warm-up stage is

$$L_{\text{seg1}} = H\left(y_i^1, M_{\text{initial}}\right) + H\left(y_i^2, M_{\text{initial}}\right) + H\left(y_i^3, M_{\text{initial}}\right) \tag{11}$$

where H represents the standard cross-entropy loss function.

In the later stage of training, we use the high-confidence pseudolabel obtained by the CNN branches to train the hybrid branch. The loss function consists of three parts: CNN segmentation loss, hybrid branch segmentation loss, and knowledge distillation loss. Therefore, the segmentation loss function is

$$L_{\text{seg2}} = H\left(y_i^1, M_{\text{initial}}\right) + H\left(y_i^2, M_{\text{initial}}\right) + H\left(y_i^3, M_{\text{pse}}\right) + \text{Dice}\left(y_i^3, M_{\text{pse}}\right). \tag{12}$$

Dice represents the Dice loss function [54]. According to the above, the SparseFormer loss function can be expressed as follows:

$$L_{\text{total}} = \begin{cases} L_{\text{seg1}} + \beta \times L_{\text{kd}}, & \text{if } t < N \\ L_{\text{seg2}} + \beta \times L_{\text{kd}}, & \text{else} \end{cases} \tag{13}$$

β is the weight value of the balanced knowledge distillation loss function, L_{kd} represents the knowledge distillation loss function, t refers to the number of current training epochs, and N is the total number of epochs in the warm-up stage.

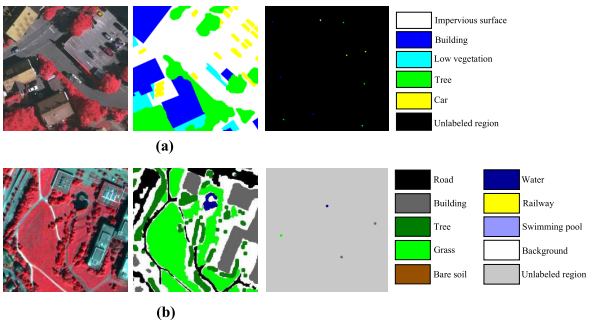


Fig. 5. Illustration of (a) Vaihingen and (b) Zurich summer datasets.

IV. Experiments

A. Datasets

1) Vaihingen Dataset: This dataset consists of 33 aerial images with a resolution of 9 cm, averaging approximately 2500 × 1900 pixels, covering a total area of about 1.38 km^2 [Fig. 5(a)]. The images include near-infrared, red, and green bands. As for the Vaihingen dataset, 16 are annotated with six land cover types: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. To ensure fairness in experiments, we select 11 images as training data (IDs: 1, 3, 5, 7, 13, 17, 21, 23, 26, 32, and 37) based on the training–testing split scheme used in the previous research, with the remaining five images (IDs: 11, 15, 28, 30, and 34) using as the test set.

2) Zurich Summer Dataset: This dataset comprises 20 satellite images captured by the QuickBird satellite in Zurich in August 2002 [Fig. 5(b)]. Each image has a spatial resolution of 0.62 m and an average size of 1000 × 1000 pixels, containing four channels: near-infrared, red, green, and blue. Only the near-infrared, red, and green channels are utilized for the experiments, and five images (IDs: 16, 17, 18, 19, and 20) are selected as the test set, while the remaining 15 images are used for training. The dataset includes eight urban classes: road, building, tree, grass, bare soil, water, railway, and swimming pool, with unclassified pixels marked as background. During the training phase, we use the sparse point annotations provided in previous research [20], consisting of four groups of point annotations. In the testing phase, the evaluation is performed using the complete pixelwise annotations from the original dataset.

B. Comparison Methods and Experimental Details

1) Comparison Methods: We use the FS SparseFormer method as the baseline method. To demonstrate the competitiveness of the proposed approach, we selected five sparse supervision models, including CRGNet [24], DBFNet [22], AGMM [45], SASFormer [21], and ScibeFormer [23]. In addition, we also include three FS semantic segmentation methods: FCN [36], SegFormer [29], and CMTFNet [40] for comparison.

2) Implementation Details: During the training process, we expand the training set by applying data augmentation techniques, including image rotation and vertical and horizontal

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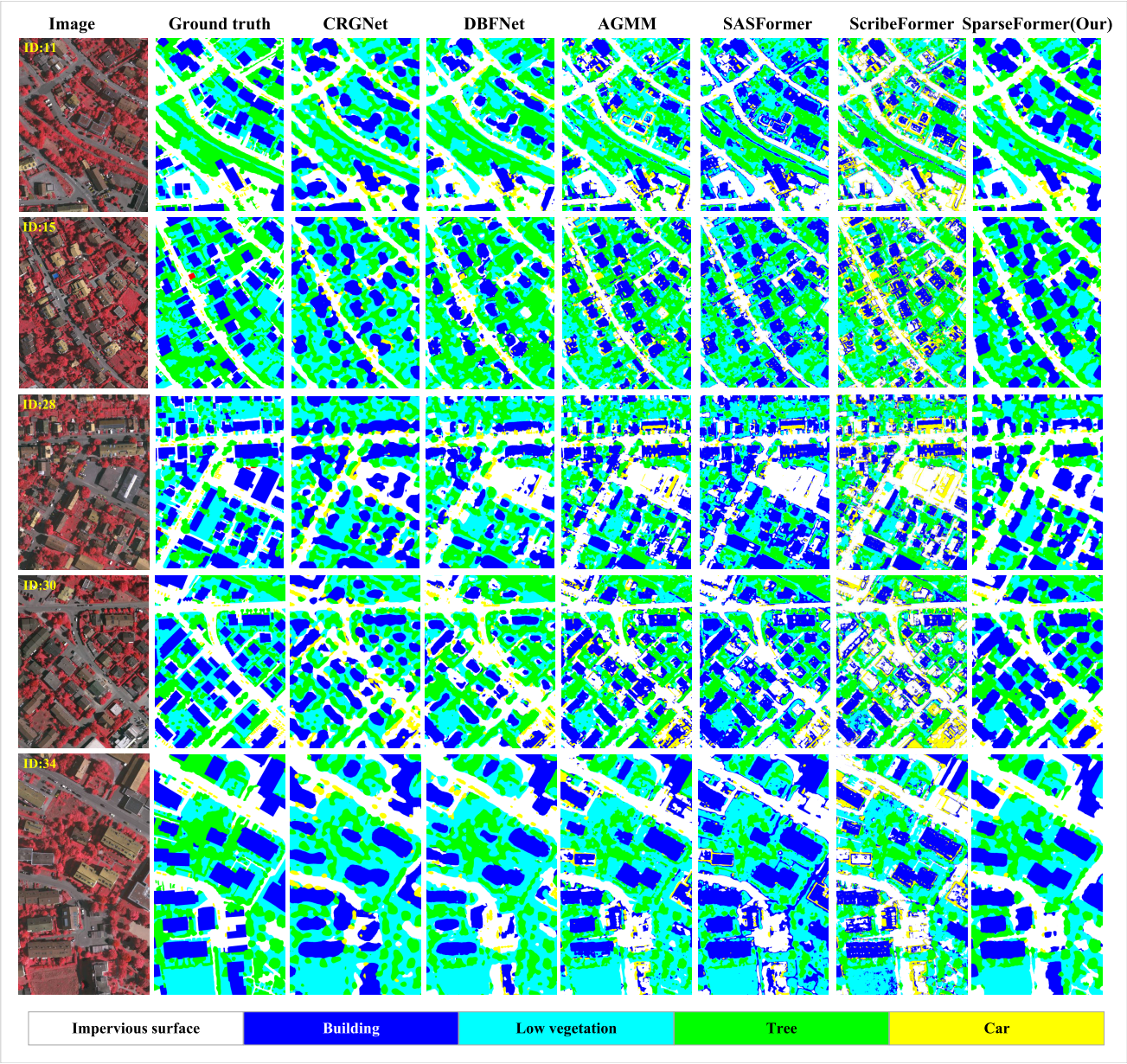


Fig. 6. Visual comparison of results of the Vaihingen dataset.

flipping. The input size for SparseFormer is 512×512, and we use the AdamW optimizer with a learning rate of 0.0001 and a batch size of 2, training for 50 epochs with the parameter β was set to 0.3 and T was set to 10. In the testing phase, we crop the images into 512×512 patches with a stride of 100 pixels. Finally, these segmentation maps are concatenated to create a complete semantic segmentation map, which is evaluated by the ground truth. The evaluation metrics included the F_1 score and mean intersection over union (mIoU). After every two epochs, we assess the accuracy of the test set, and the result is taken as the best results from all epochs. Each experiment is conducted on four training sets provided by Hua et al. [20] to compute the fluctuation accuracy value. All experiments are completed using the Pytorch framework on a 16-GB memory Tesla A100 GPU, ensuring consistent parameter settings for reasonable and fair comparisons.

C. Experimental Results

Tables I and II summarize the quantitative evaluation results on the Vaihingen and Zurich Summer datasets. For the Vaihingen dataset, the proposed method achieves an mF_1 score of 70.32% and an mIoU of 55.67%, with F_1 scores for all categories surpassing those of other WS methods. For the Zurich Summer dataset, SparseFormer achieves an mF_1 score of 75.07% and an mIoU of 64.85%, with overall accuracy higher than other WS approaches. Notably, SparseFormer exhibits significant advantages in identifying roads, buildings, trees, and water bodies in the Zurich Summer dataset.

Previous sparse supervision methods often lack mechanisms to control the quality of pseudolabels, leading to error propagation during training. In contrast, the proposed method not only evaluates the confidence of pseudolabels but also strategically integrates the strengths of multiple models, substantially improving the overall quality of pseudolabels. Furthermore,

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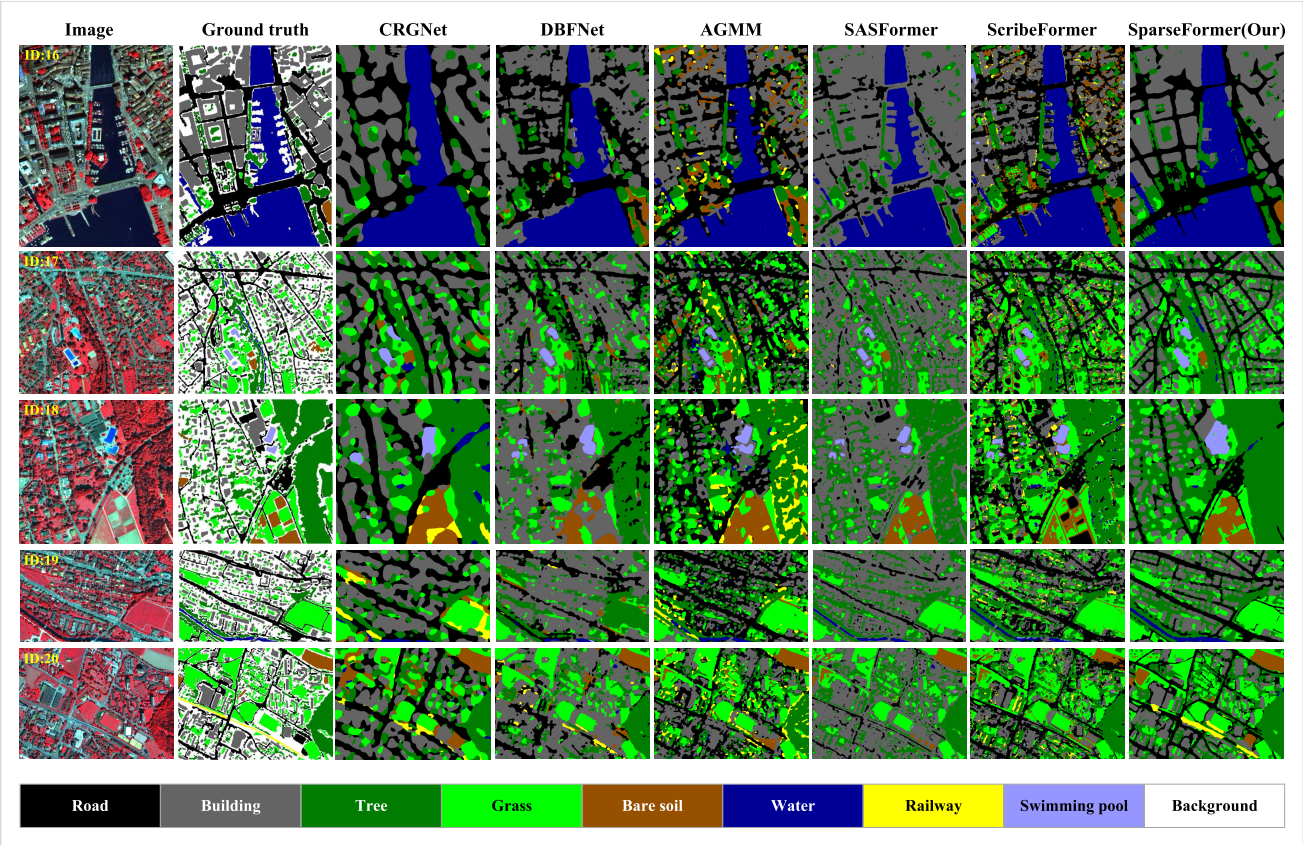


Fig. 7. Visual comparison of classification results of the Zuirch Summer dataset.

TABLE I

EXPERIMENTAL RESULTS OF THE VAIHINGEN DATASET (%). THE PER-CLASS F_1 SCORE, mF_1 SCORE, AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

	Method	Impervious Surface	Building	Low vegetation	Tree	Car	mF_1	mIoU
FS	SparseFormer (Hybrid)	89.39	93.66	76.64	85.42	81.57	85.33	74.89
	SparseFormer (CA)	88.03	93.19	74.36	83.94	76.41	83.19	71.84
	SparseFormer (CBAM)	88.35	93.54	73.45	83.82	77.96	83.43	72.21
	FCN	87.6	92.19	75.15	84.41	78.65	83.6	72.29
	SegFormer	87.73	92.85	76.26	85.54	80.56	84.59	73.72
	CMTFNet	89.05	93.89	76.87	85.23	83.37	85.68	75.38
WS	CRGNet	67.65±2.86	78.3±1.73	57.86±1.34	72.6±3.91	36.38±3.43	62.56±1.75	47.12±1.96
	DBFNet	67.88±4.2	70.55±4.92	53.1±2.52	74.43±2.36	35.57±12.43	60.31±2.84	44.76±2.27
	AGMM	70.58±0.71	72.77±3.06	57.73±1.56	75.76±2.01	21.11±5.04	59.59±1.1	45.05±0.9
	SASFormer	67.18±4.18	68.36±1.27	57.34±1.61	74.31±3.6	37.88±11.45	61.01±2.96	45.18±2.61
	ScribeFormer	70.04±3.98	73.7±3.8	51.65±4.91	73.94±2.93	16.61±1.32	57.19±2.98	43.03±3.3
	SparseFormer	**78.54±1.52**	**83.63±0.27**	**59.1±5.21**	**78.36±0.83**	**51.96±6.39**	**70.32±0.88**	**55.67±0.97**

unlike traditional approaches that rely on dCRF for postprocessing, SparseFormer adaptively refines pseudolabels during training, enabling the model to capture high-resolution semantic scene details better and improve generalization capability. SparseFormer achieves significantly higher extraction accuracy for Car in the Vaihingen dataset than other WS models. This improvement is attributed to its ability to capture fine-grained details while effectively modeling global context.

However, SparseFormer performs poorly on grassland, bare soil, and swimming pools in the Zurich Summer dataset, F_1 of which are slightly lower than that of other methods. This could be due to the less distinctive features of grassland and bare soil and the small proportion of swimming pools in the images, which affects the quality of pseudolabels. In comparison, other regularization-based methods (e.g., DBFNet and SASFormer) enforce feature consistency in regions with similar characteristics, leading to higher sensitivity to these categories. Nevertheless, these WS methods remain less effective than SparseFormer in other categories and overall performance. In conclusion, the proposed SparseFormer demonstrates

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TABLE II

EXPERIMENTAL RESULTS OF THE ZURICH SUMMER DATASET (%). THE PER-CLASS F_1 SCORE, mF_1 SCORE, AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

	Method	Road	Building	Tree	Grass	Bare soil	Water	Railway	Swimming	mF_1	mIoU
FS	SparseFormer (Hybrid)	91.33	94.34	94.27	91.53	69.76	97.91	60.12	75.36	84.33	74.97
	SparseFormer (CA)	89.56	92.4	94.31	91.7	72.38	97.4	31.86	89.26	82.36	74.01
	SparseFormer (CBAM)	89.6	92.59	94.07	91.28	70.3	97.22	41.58	90.54	83.4	74.74
	FCN	88.35	91.63	93.09	88.57	59.86	96.8	58.38	92.55	83.78	74.47
	SegFormer	90.09	93.37	93.69	92.43	77.05	96.25	45.91	82.23	83.88	74.83
	CMTFNet	91.61	94.4	94.61	92.29	73.88	98.18	40.99	87.16	84.14	75.92
WS	CRGNet	67.82±4.7	75.54±3.5	77.65±2.16	70.16±7.83	52.15±8.1	89.53±1.88	6.17±4.29	70.66±10.4	63.71±3.78	50.68±4.22
	DBFNet	65.14±2.39	66.93±5.26	75.15±3.34	74.96±5.72	64.76±8.23	92.46±2.13	**20.58±16.02**	81.72±8.38	67.71±2.94	54.43±3.8
	AGMM	66.02±7.98	71.53±2.75	84.63±2.12	**85.99±2.16**	57.73±2.22	89.36±2.75	0.02±0.04	**87.18±11.52**	67.81±2.43	56.79±3.31
	SASFormer	61.79±16.69	72.55±6.76	84.06±1.87	79.1±3.45	**67.84±4.12**	94.3±1.17	2.48±4.86	84.87±5.76	68.37±2.19	57.21±2.7
	ScribeFormer	67.81±4.92	72.49±7.94	82.87±2.74	80.62±2.47	53.18±7.59	92.05±1.35	1.07±1.26	64.41±32.98	64.31±6.62	52.92±7.35
	SparseFormer	**78.42±4.67**	**84.19±3.79**	**88.89±1.73**	84.52±3.85	66.84±9.7	**96.31±0.72**	17.57±28.03	83.78±4.52	**75.07±4.99**	**64.85±4.79**

TABLE III

EVALUATION RESULTS OF LOSS ABLATION ACCURACY (%). THE mF_1 SCORE AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

$L_{dice+ce}$	L_{ce}	PAR	L_{kd}	Vaihingen		Zurich Summer	
				mF_1	mIoU	mF_1	mIoU
	✓			64.01±1.41	49.79±1.56	66.51±7.56	54.97±7.22
✓				65.39±2.79	50.82±3.11	72.73±5.38	60.98±5.88
✓			✓	65.53±2.42	50.85±1.94	70.13±3.01	58.47±4.2
✓		✓		68.38±3.76	53.89±3.4	74.62±2.52	64.77±2.69
	✓	✓	✓	69.69±1.35	54.69±1.43	74.21±3.3	63.39±3.47
*		*	*	69.71±1.96	55.25±1.77	74.38±4.78	63.55±5.14
✓		✓	✓	**70.32±0.88**	**55.67±0.97**	**75.07±4.99**	**64.85±4.79**

* denotes training without credible assessment.

TABLE IV

RESULTS OF ATTENTIONAL MECHANISM ABLATION EXPERIMENT (%). THE mF_1 SCORE AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

CNN1		CNN2		Vaihingen		Zurich Summer	
CA	CBAM	CA	CBAM	mF_1	mIoU	mF_1	mIoU
✓		✓		69.83±1.02	55.32±1	72.65±4.66	61.73±5.28
	✓		✓	69.6±0.96	54.94±0.95	73.17±5.05	62.58±5.66
✓			✓	**70.32±0.88**	**55.67±0.97**	**75.07±4.99**	**64.85±4.79**

significant advantages over existing sparse supervision-based WS methods, highlighting its potential for high-resolution RS image semantic segmentation.

We show the full-image qualitative result of the test set on both datasets in Figs. 6 and 7. It can be observed that results produced by SparseFormer are highly consistent with the ground-truth values. Compared with other WS classification methods, our method shows more accurate ground object boundaries and higher local consistency because the PAR module makes full use of the underlying information of the image and adaptively enhances the probability of CNN outputs.

We present detailed comparison images of the two datasets in Fig. 8 to visually compare the semantic segmentation results. The results show that SparseFormer excels in capturing land cover details and boundary information. For instance, in the Vaihingen dataset, SparseFormer accurately identifies the car category, showing high agreement with ground truth. Despite limited annotations, SparseFormer effectively distinguishes between highly similar land covers, such as low vegetation and trees. In the Zurich Summer dataset, SparseFormer can identify small distribution categories, such as railways, which other models struggle to recognize accurately. On the other hand, results achieved by SparseFormer contain less noise, attributing to the credible assessment of the generated pseudolabels, where only the high-confidence regions are selected for training. Consequently, SparseFormer is less affected by erroneous labels during training than other WS methods, ultimately achieving cleaner results on both datasets.

D. Ablation Experiments

1) Ablation Results of Loss Function: To verify the necessity of each step in SparseFormer, we conduct corresponding ablation experiments of the loss function, with the accuracy evaluation results shown in Table III. First, training SparseFormer using point annotations with only cross-entropy loss achieves 49.79% and 54.97% mIoU on the Vaihingen and Zurich Summer datasets, respectively, which are the lowest accuracy in all experiments. The results show that the model can learn with only sparse point annotations, but the performance is limited. Fig. 9 presents the loss and training accuracy visualization curves for the three branches. It can be observed that the three branches quickly fit the point annotations, and there is no significant improvement in accuracy after about 20 epochs [Fig. 9(a) and (c)]. On the contrary, the loss of SparseFormer, which includes all modules, still decreases in the later stage and the training accuracy still increases significantly [Fig. 9(b) and (d)], indicating that each module and step designed in SparseFormer can effectively avoid the occurrence of the overfitting phenomenon, and help the hybrid branch learn new knowledge from the pseudolabel, thus improving the performance of the model.

Second, the pseudolabels generated by the two CNN models guide the learning of the hybrid branch, indicating that pseudolabels provide more semantic information than point annotations, significantly enhancing the learning quality of the hybrid branch. Since the CNN branches only use sparse

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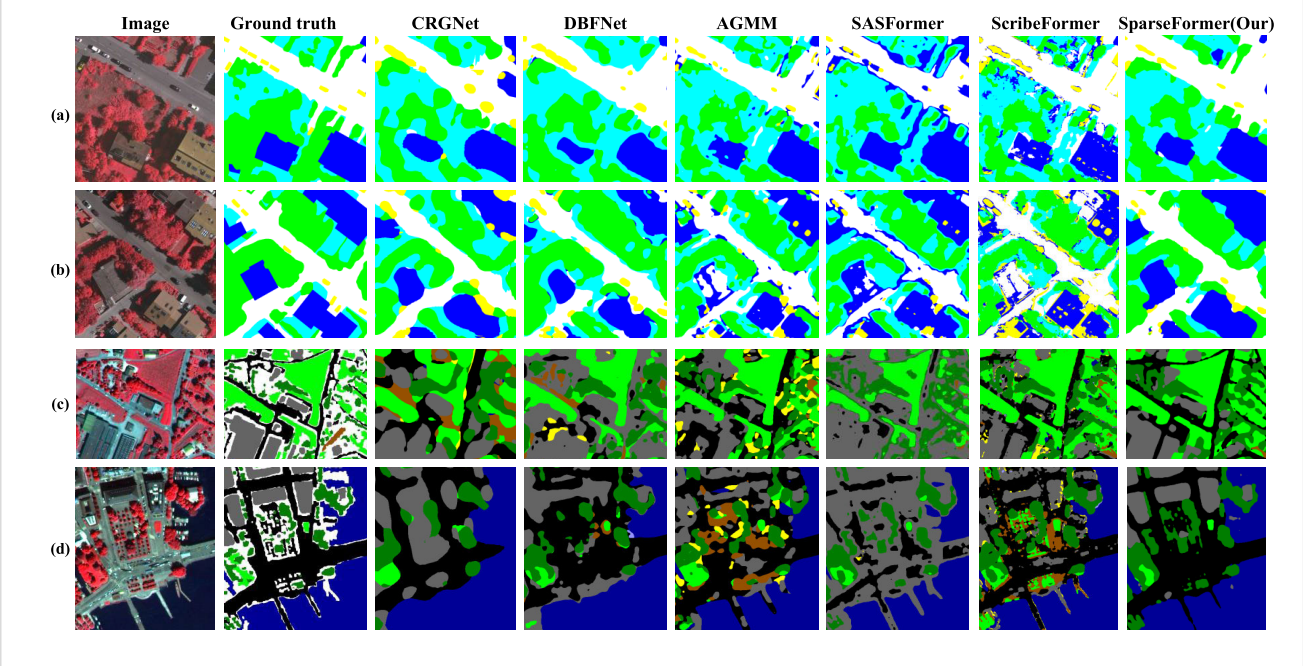


Fig. 8. Comparison of results in detail. (a) and (b) Vaihingen dataset. (c) and (d) Zurich Summer dataset.

annotations for training, they quickly fit these limited labels, which limits their learning performance in the later training stages, preventing further improvement in the quality of generated pseudolabels. The knowledge distillation module addresses this issue by transferring information from different branches to the CNN branch in the form of soft probabilities, helping the CNN branch acquire information beyond the sparse annotations, thus enabling the generation of more accurate pseudolabels and improving overall segmentation quality. In addition, the accuracy evaluation results show that the credibility assessment we designed provides a certain performance gain on both datasets. This indicates that the credibility assessment effectively controls the quality of pseudolabels and prevents the propagation of incorrect labels during the training process.

Furthermore, the performance of SparseFormer significantly improved with the addition of the PAR module, as shown in Fig. 10. This result indicates that utilizing low-level information from images to consolidate the quality of pseudolabels and enhance the local consistency of labels is essential in the sparse supervised semantic segmentation. Due to the lack of complete semantic information when CNN models handle sparse labels, it becomes challenging for the model to effectively learn details such as the contours and structures of objects with limited annotations, leading to limitations in the directly generated pseudolabels. By introducing the PAR module, we can better integrate and utilize low-level information, thereby significantly improving the generation of pseudolabels and enhancing the model's overall performance.

2) Ablation Results of Attention Mechanism: In this section, we conduct an ablation experiment of the attention mechanism, and the results are shown in Table IV. The gain is smaller when two CNN branches use the same attention mechanism. This is mainly due to the lack of significant differences between the output results, resulting in the pseudolabel failing to absorb the respective strengths of dual-CNN branches. In the case of sparse annotations, if the prediction of one CNN branch is biased, the error propagation phenomenon will be caused, and the overall performance of the hybrid branch will be further affected. In contrast, when the two CNN branches adopt different attention mechanisms, we achieve the highest accuracy in this article. This finding suggests that pseudolabels generated under credible assessment have higher quality, thus better supporting the learning process. The model can better capture the diversity and importance between features by introducing different attention mechanisms, enhancing adaptability to complex scenes [55].

Fig. 11 illustrates the uncertainty maps generated by two CNN branches under different attention mechanisms. It can be observed that uncertainty is mainly consistent in most areas, which are regions where the model can easily identify objects under sparse annotations. The differences primarily occur at object boundaries or in areas, where classes are challenging to distinguish. Different attention mechanisms can help uncover these regions that standard models struggle to recognize, thereby improving the quality of the pseudolabels. Therefore, the proposed dual-CNN strategy with different attention mechanisms improves overall performance and segmentation accuracy under sparse supervision, highlighting the importance of diverse feature learning.

3) Parameter Analysis: In SparseFormer, the only parameter that needs to be manually set is the weight of the knowledge distillation loss function. To investigate the impact of this parameter on model performance, we conducted detailed ablation experiments, setting probability values to 0, 0.1, 0.3, and 0.5. Tables V and VI show that high classification accuracy is achieved in most cases when $\beta = 0.3$. The experimental results indicate that the knowledge distillation

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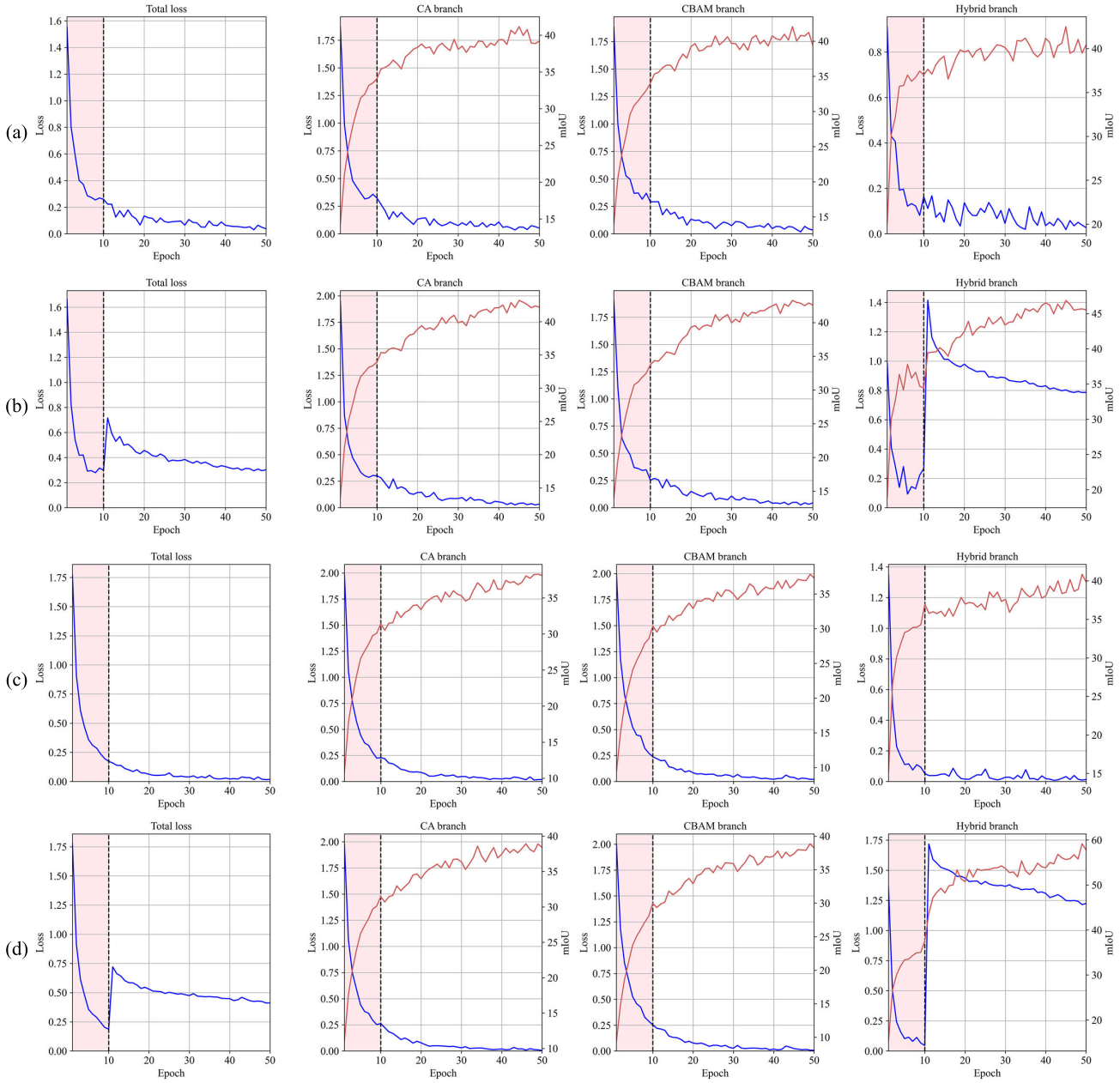


Fig. 9. Cross-entropy loss function ablation experiments of different branches, where blue represents the loss function, red represents the training accuracy, and the first ten epochs is the warm-up stage. (a) Results of Vaihingen dataset using only cross-entropy loss function. (b) Results of Vaihingen dataset using all modules. (c) Results of Zurich Summer dataset using only cross-entropy loss function. (d) Results of Zurich Summer dataset using all modules.

strategy generally provides accuracy gains for the model on both datasets. This suggests that the knowledge learned in the hybrid branch effectively feeds back to the CNN branch, improving the quality of the pseudolabels, thereby enhancing overall classification accuracy. The knowledge distillation enhances the model's perception of the category with a small distribution ("Car" in the Vaihingen dataset and "Railway" in the Zurich Summer dataset) by transferring fine-grained feature information learned by the hybrid branch to the CNN branch, compensating for the CNN's shortcomings in handling these areas.

However, we also observe that in some cases, knowledge distillation led to a decrease in classification accuracy. This may be due to the uncertainty of soft probabilities during the distillation process, which can feed back some erroneous information from the hybrid branch to the CNN branch. These errors may be further amplified during the pseudolabel generation process, leading to cumulative errors in the training process and ultimately affecting the model performance. This situation indicates that while knowledge distillation is beneficial in most cases, it still carries certain risks in specific scenarios, mainly when the model's judgment of low-confidence areas is imprecise, potentially exacerbating error propagation.

Despite some limitations, the knowledge distillation operation in SparseFormer demonstrates significant advantages in sparse annotation scenarios. It provides the CNN branch with diverse and rich information inputs, especially in cases of point

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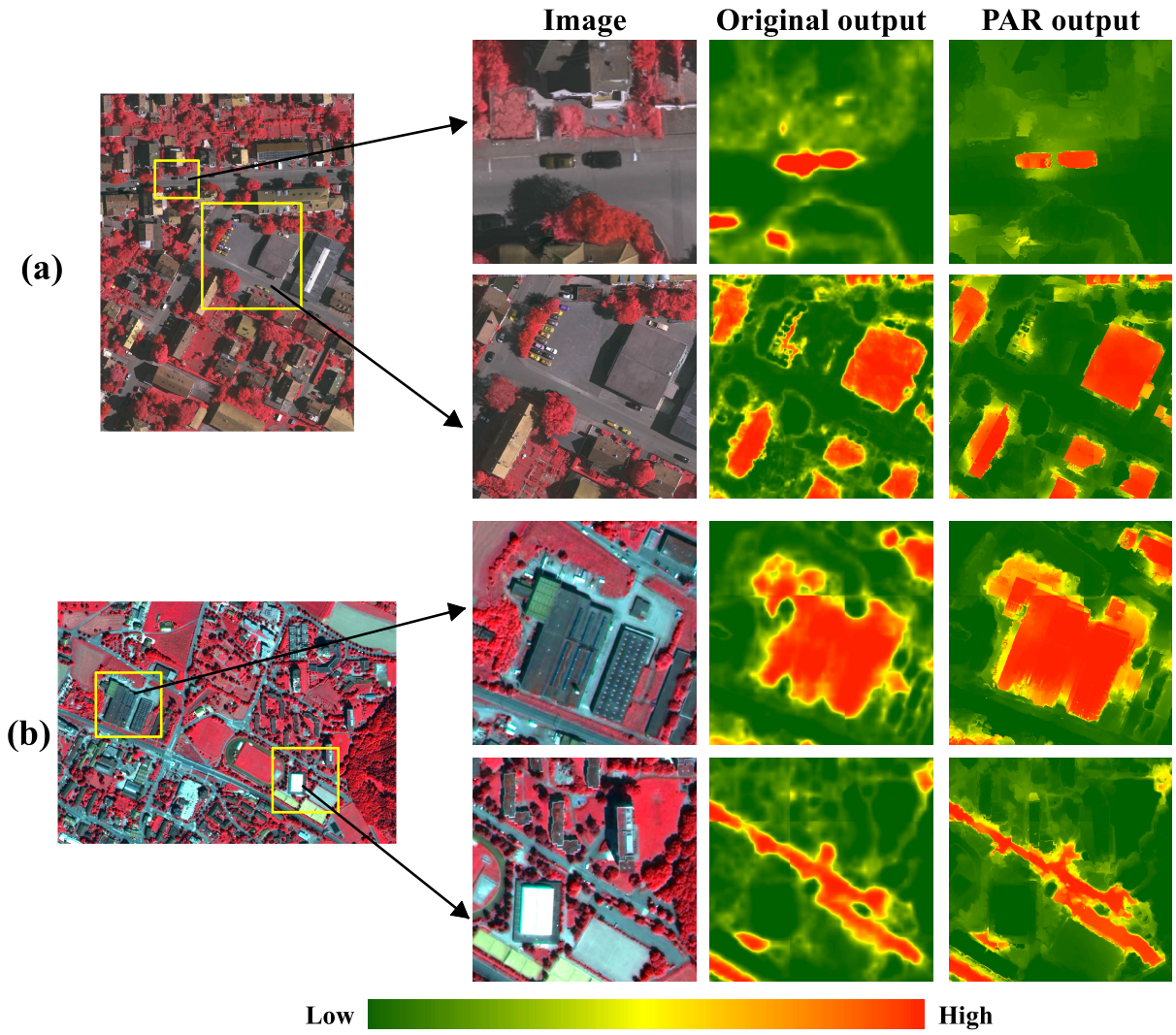


Fig. 10. Visualization of PAR module effect. (a) Vaihingen dataset. (b) Zurich Summer dataset.

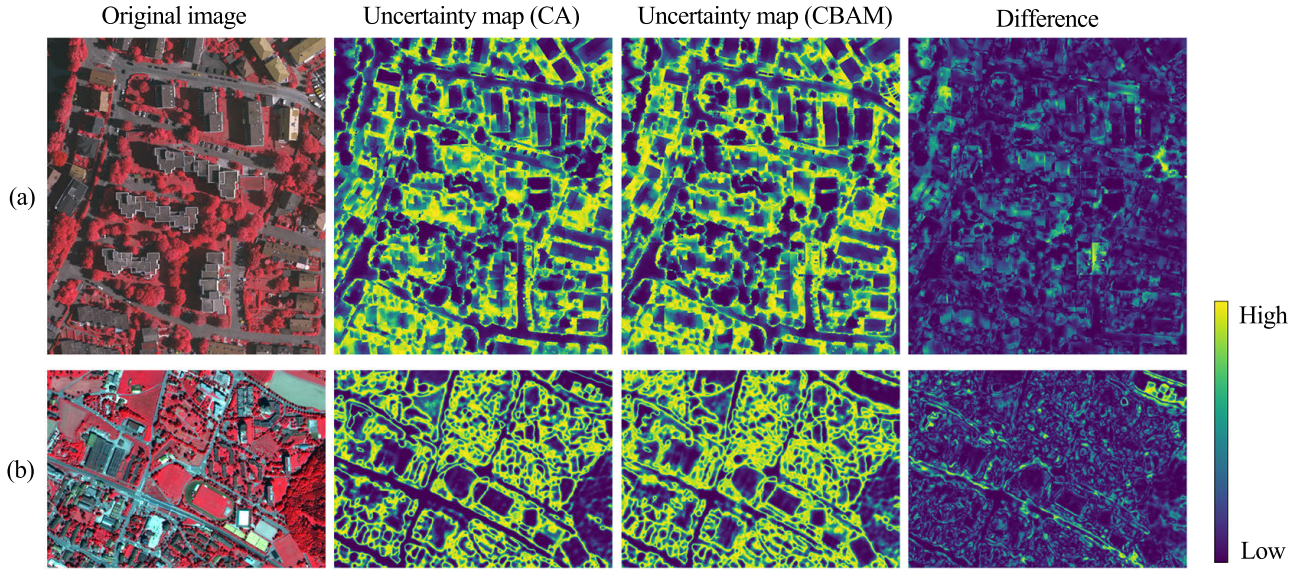


Fig. 11. Visualization of segmentation uncertainty for different attention mechanisms. (a) Vaihingen dataset. (b) Zurich Summer dataset.

annotated data, preventing the overfitting phenomenon caused by excessive reliance on the limited label data. This enables the model to improve pseudolabel quality while sustaining high classification accuracy and stability.

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TABLE V
ANALYSIS RESULTS OF PARAMETER β IN VAIHINGEN DATASET (%). THE mF_1 SCORE AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

β	Impervious Surface	Building	Low vegetation	Tree	Car	mF_1	mIoU
0	77.2±1.22	83.3±2.28	60.57±5.27	77.28±2.67	43.46±12.73	68.38±3.76	53.89±3.4
0.1	**79.08±1.87**	**84.81±1.65**	**60.58±3.28**	**79.03±0.53**	46.75±9.07	70.05±2.43	**55.75±2.37**
0.3	78.54±1.52	83.63±0.27	59.1±5.21	78.36±0.83	51.96±6.39	**70.32±0.88**	55.67±0.97
0.5	78.26±2.65	83.73±0.94	55.87±6.32	78.38±0.79	**54.5±7.13**	70.11±2.3	55.44±2.36

TABLE VI
ANALYSIS RESULTS OF PARAMETER β IN ZURICH SUMMER DATASET (%). THE mF_1 SCORE AND MIOU. BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

β	Road	Building	Tree	Grass	Bare soil	Water	Railway	Swimming	mF_1	mIoU
0	**79.65±3.55**	84.64±3.15	**88.96±1.5**	**86.19±1.96**	**75.2±9.26**	95.9±0.69	7.3±9.47	79.09±3.42	74.62±2.52	64.77±2.69
0.1	79.3±4.12	**84.73±2.23**	86.25±3.98	81.62±5.21	70.45±7.87	95.01±1.58	13.7±15.46	83.25±5.57	74.29±3.9	63.68±4.19
0.3	78.42±4.67	84.19±3.79	88.89±1.73	84.52±3.85	66.84±9.7	**96.31±0.72**	**17.57±28.03**	83.78±4.52	**75.07±4.99**	64.85±4.79
0.5	76.61±4.93	83.77±2.36	86.9±2.45	83.91±2.31	72.59±6.56	94.58±4.05	9.06±10.75	**84.33±6.45**	73.97±2.65	**66.41±5.69**

E. Advantages and Limitations of SparseFormer

The advantages of SparseFormer are reflected in two main aspects. First, the proposed method enables end-to-end training, differing from other approaches that rely on a single model or technique to generate pseudolabels; instead, SparseFormer integrates the strengths of two CNN models. This integration enhances the quality of pseudolabels, reducing the negative impact of uncertainty on model training, thus improving training efficiency and adaptability to complex scenarios. Second, our method effectively combines the advantages of CNNs and Transformers. The local information localization ability of CNNs ensures accurate detail capture, while the global feature modeling capability of Transformers aids in understanding the relationships between different regions of the image. By establishing a hybrid branch, we can delve into the adequate information in pseudolabels, preventing the model from overfitting to its generated pseudolabels and improving generalization ability. Although SparseFormer has achieved satisfactory results on two datasets, it also has limitations. For example, point annotations may fail to accurately reflect land cover boundaries and semantic relationships, leading to discontinuities in segmentation results. The future research will optimize pseudolabel generation and feature extraction to address boundary recognition and semantic coherence issues under sparse point supervision.

V. CONCLUSION

This article presents SparseFormer, a credible dual-CNN expert-guided Transformer model designed for RS semantic segmentation using point-level annotations. SparseFormer comprises three branches, where two CNN branches employ different attention mechanisms to promote diverse outputs. To improve the local consistency of pseudolabels, we incorporate a PAR module that dynamically refines CNN output probabilities by integrating image information. Subsequently, we propose a credible assessment mechanism to mix the CNN outputs, resulting in high-quality pseudolabels dynamically. The hybrid branch, trained on these pseudolabels, effectively captures additional contextual information by combining the strengths of CNNs in local feature extraction with the long-range context modeling capabilities of Transformers. Furthermore, we introduce a knowledge distillation strategy to facilitate knowledge sharing between branches, enhancing the overall performance of the model. Our extensive experiments on two benchmark datasets demonstrate that SparseFormer significantly outperforms state-of-the-art methods, effectively reducing annotation burdens while achieving high-quality results. The future research will focus on controlling and optimizing the quality of pseudolabels and exploring the learning mechanisms of image features under point supervision.

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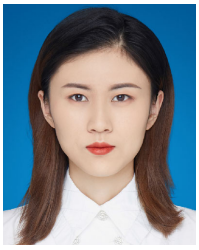
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